## II. Write brief answer to the following questions:

26. State Boyle's law.
27. Name two items that can serve as a model for Gay Lusaac' law and explain.
28. Give the mathematical expression that relates gas volume and moles.
29. What are ideal gases? In what way real gases differ from ideal gases.
30. Can a Van der Waals gas with a=0 be liquefied? explain.
31. Suppose there is a tiny sticky area on the wall of a container of gas. Molecules hitting this area stick there permanently. Is the pressure greater or less than on the ordinary area of walls?
32. Explain the following observations
    a) Aerated water bottles are kept under water during summer
    b) Liquid ammonia bottle is cooled before opening the seal
    c) The tyre of an automobile is inflated to slightly lesser pressure in summer than in winter
    d) The size of a weather balloon becomes larger and larger as it ascends up into larger altitude
33. Give suitable explanation for the following facts about gases.
    a) Gases don't settle at the bottom of a container
    b) Gases diffuse through all the space available to them
34. Suggest why there is no hydrogen ($H_2$) in our atmosphere. Why does the moon have no atmosphere?
35. Explain whether a gas approaches ideal behavior or deviates from ideal behaviour if
    a) it is compressed to a smaller volume at constant temperature.
    b) the temperature is raised while keeping the volume constant
    c) more gas is introduced into the same volume and at the same temperature
36. Which of the following gases would you expect to deviate from ideal behaviour under conditions of low temperature $F_2$, $Cl_2$ or $Br_2$? Explain.
37. Distinguish between diffusion and effusion.
38. Aerosol cans carry clear warning of heating of the can. Why?
39. Would it be easier to drink water with a straw on the top of Mount Everest?
40. Write the Van der Waals equation for a real gas. Explain the correction term for pressure and volume
41. Derive the values of critical constants in terms of van der Waals constants.
42. Why do astronauts have to wear protective suits when they are on the surface of moon?
43. When ammonia combines with HCl, $NH_4Cl$ is formed as white dense fumes. Why do more fumes appear near HCl?



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